

Use Case Template — Upload Study Material

1. Use Case Title	Upload Study Material
2. Abbreviated Title	Upload Material
3. Use Case ID	UC-01
4. Actors	Student
5. Description	
The student uploads study material such as PDFs or notes. Study Buddy processes the material and makes it available to the Teacher Agent as the ground truth for teaching and evaluation.	
5.1. Pre-Conditions	
1. Student is logged in. 2. A supported study material file is available. 3. The upload interface is accessible.	
5.2. Task Sequence	
1. Student selects a study material file. 2. System receives and validates the uploaded file. 3. System extracts the document content. 4. Teacher Agent processes the material as the ground truth. 5. System confirms that the material is ready for the learning session.	
5.3. Post-Conditions	
1. Study material is stored and processed. 2. Teacher Agent has access to the prepared source material. 3. The material is available for teaching and later evaluation.	
6. Modification History	Date: 31-Aug-2026
7. Author	Study Buddy Team

Use Case Template — Conduct Cold Baseline Evaluation

1. Use Case Title	Conduct Cold Baseline Evaluation
2. Abbreviated Title	Cold Baseline
3. Use Case ID	UC-02
4. Actors	Student
5. Description	
Study Buddy evaluates the Child Agent before teaching to measure its pre-existing knowledge and establish a baseline for comparison with the post-teaching performance.	
5.1. Pre-Conditions	
1. Study material has been uploaded and processed. 2. A baseline evaluation can be generated. 3. Child Agent has not received the student's teaching-derived memory for the evaluation.	
5.2. Task Sequence	
1. Student starts the baseline evaluation. 2. Teacher Agent generates questions from the source material. 3. System presents the questions for the Child Agent to answer. 4. Child Agent answers using its available knowledge. 5. Teacher Agent evaluates the answers against the source material. 6. System records the baseline score.	
5.3. Post-Conditions	
1. Cold baseline score is recorded. 2. Baseline results are available for later comparison. 3. Potential pretrained knowledge leakage can be assessed using the baseline.	
6. Modification History	Date: 31-Aug-2026
7. Author	Study Buddy Team

Use Case Template — Teach Child Agent

1. Use Case Title	Teach Child Agent
2. Abbreviated Title	Teach Student Model
3. Use Case ID	UC-03
4. Actors	Student
5. Description	
The student teaches the Child Agent a selected topic. The teaching interaction provides the information from which the system constructs the Child Agent's structured knowledge.	
5.1. Pre-Conditions	
1. Study material has been uploaded and processed. 2. A teaching session has been started. 3. The Child Agent is isolated from the source material.	
5.2. Task Sequence	
1. Student starts a teaching session. 2. System provides the teaching interface. 3. Student explains the topic to the Child Agent. 4. Child Agent responds during the interaction. 5. System records the teaching interaction for memory extraction. 6. System completes the teaching session.	
5.3. Post-Conditions	
1. Teaching interaction has been completed. 2. The interaction is available to the memory and learning engine. 3. The Child Agent remains separated from the original source material.	
6. Modification History	Date: 31-Aug-2026
7. Author	Study Buddy Team

Use Case Template — Teach via Text

1. Use Case Title	Teach via Text
2. Abbreviated Title	Text Teaching
3. Use Case ID	UC-04
4. Actors	Student
5. Description	
The student teaches the Child Agent through a text-based conversation. The conversation is used as the input to the memory construction process.	
5.1. Pre-Conditions	
1. A teaching session has been started. 2. The text teaching interface is available. 3. Child Agent cannot access the original study material.	
5.2. Task Sequence	
1. System displays the text conversation interface. 2. Student enters an explanation or response. 3. System sends the text to the Child Agent. 4. Child Agent responds to the student. 5. Student continues explaining or clarifying the topic. 6. System passes the completed interaction to the memory and learning engine.	
5.3. Post-Conditions	
1. Text teaching interaction is completed. 2. Teaching content is available for structured memory extraction. 3. Child Agent has not accessed the original source material.	
6. Modification History	Date: 31-Aug-2026
7. Author	Study Buddy Team

Use Case Template — Teach via Voice

1. Use Case Title	Teach via Voice
2. Abbreviated Title	Voice Teaching
3. Use Case ID	UC-05
4. Actors	Student
5. Description	
The student teaches the Child Agent through spoken interaction. Speech is converted to text and supplied to the learning engine for structured memory extraction.	
5.1. Pre-Conditions	
1. A teaching session has been started. 2. Voice input and speech-to-text processing are available. 3. Child Agent cannot access the original study material.	
5.2. Task Sequence	
1. System activates the voice teaching interface. 2. Student explains the topic verbally. 3. System converts the student's speech into text. 4. System sends the transcribed teaching interaction to the Child Agent. 5. Child Agent responds to the student. 6. System passes the interaction to the memory and learning engine.	
5.3. Post-Conditions	
1. Voice teaching interaction is completed. 2. A text representation of the teaching session is available for memory extraction. 3. Child Agent remains separated from the original source material.	
6. Modification History	Date: 31-Aug-2026
7. Author	Study Buddy Team

Use Case Template — Extract & Update Memory

1. Use Case Title	Extract & Update Memory
2. Abbreviated Title	Memory Update
3. Use Case ID	UC-06
4. Actors	Student
5. Description	
The learning engine converts the teaching interaction into structured knowledge containing concepts, dependencies, and confidence information, and updates the Child Agent's memory.	
5.1. Pre-Conditions	
1. A teaching interaction has been completed. 2. The teaching interaction is available to the learning engine. 3. The source material remains inaccessible to the Child Agent.	
5.2. Task Sequence	
1. System receives the teaching interaction. 2. Learning engine extracts concepts and definitions from the interaction. 3. Learning engine identifies relevant dependencies or relationships. 4. Learning engine assigns a rough confidence score. 5. System stores the structured knowledge in the Child Agent's memory. 6. System discards the original conversation after the structured memory is created.	
5.3. Post-Conditions	
1. Child Agent's structured memory is updated. 2. The stored memory contains extracted concepts and confidence information. 3. The original conversation is no longer used as the Child Agent's knowledge source.	
6. Modification History	Date: 31-Aug-2026
7. Author	Study Buddy Team

Use Case Template — Take Closed-Book Exam

1. Use Case Title	Take Closed-Book Exam
2. Abbreviated Title	Closed-Book Exam
3. Use Case ID	UC-07
4. Actors	Student
5. Description	
The Child Agent takes an exam generated from the original study material after teaching, while answering using only its structured memory from the teaching session.	
5.1. Pre-Conditions	
1. Study material has been uploaded and processed as well as teaching session and memory construction are complete. 2. An exam can be generated from the source material and the Child Agent is restricted to its structured memory.	
5.2. Task Sequence	
1. Student starts the closed-book evaluation. 2. Teacher Agent generates an exam from the source material. 3. System presents the generated questions to the Child Agent. 4. Child Agent answers the questions using only its structured memory. 5. System submits the answers for evaluation. 6. Teacher Agent evaluates the answers against the source material.	
5.3. Post-Conditions	
1. Child Agent's exam answers are recorded. 2. Teacher Agent has evaluated the answers. 3. A Teaching Efficacy Score can be calculated from the exam performance.	
6. Modification History	Date: 31-Aug-2026
7. Author	Study Buddy Team

Use Case Template — Grade & Evaluate Answers

1. Use Case Title	Grade & Evaluate Answers
2. Abbreviated Title	Evaluate Answers
3. Use Case ID	UC-08
4. Actors	Student
5. Description	
The Teacher Agent evaluates the Child Agent's exam answers against the uploaded study material and produces an objective performance result.	
5.1. Pre-Conditions	
1. A closed-book exam has been completed. 2. Child Agent's answers are available. 3. Teacher Agent has access to the source material.	
5.2. Task Sequence	
1. System sends the Child Agent's answers to the Teacher Agent. 2. Teacher Agent compares each answer with the source material. 3. Teacher Agent assigns scores or evaluation results. 4. System calculates the overall exam performance. 5. System records the Teaching Efficacy Score. 6. System prepares the evaluation result for the student.	
5.3. Post-Conditions	
1. Exam answers have been graded. 2. Teaching Efficacy Score is recorded. 3. Evaluation results are available to the student.	
6. Modification History	Date: 31-Aug-2026
7. Author	Study Buddy Team

Use Case Template — View Learning Results

1. Use Case Title	View Learning Results
2. Abbreviated Title	View Results
3. Use Case ID	UC-09
4. Actors	Student
5. Description	
The student views the results of the learning session, including the Child Agent's exam performance and the resulting Teaching Efficacy Score.	
5.1. Pre-Conditions	
1. At least one evaluation session has been completed. 2. Evaluation results have been recorded.	
5.2. Task Sequence	
1. Student opens the results section. 2. System retrieves the recorded evaluation results. 3. System displays the exam score and Teaching Efficacy Score. 4. System displays relevant learning or memory information available for the session. 5. Student reviews the results.	
5.3. Post-Conditions	
1. Student has viewed the evaluation results. 2. Session results remain recorded for later analysis.	
6. Modification History	Date: 31-Aug-2026
7. Author	Study Buddy Team